

Reproducibility Checklist

Instructions for Authors:

This document outlines key aspects for assessing reproducibility. Please provide your input by editing this .tex file directly.

For each question (that applies), replace the “Type your response here” text with your answer.

Example: If a question appears as

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
Type your response here
```

you would change it to:

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
yes
```

Please make sure to:

- Replace **ONLY** the “Type your response here” text and nothing else.
- Use one of the options listed for that question (e.g., **yes**, **no**, **partial**, or **NA**).
- **Not** modify any other part of the `\question` command or any other lines in this document.

You can `\input` this .tex file right before `\end{document}` of your main file or compile it as a stand-alone document. Check the instructions on your conference’s website to see if you will be asked to provide this checklist with your paper or separately.

1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) **partial**. The Method section gives equations for the low-pass branch, high-pass residual branch, and node-adaptive gate; explicit pseudocode is not yet included.
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) **yes**. Hypotheses and bounded evidence are separated from measured results in the abstract, result discussion, and limitations.
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) **yes**. The related-work section cites standard GNN, heterophily, polynomial-filter, and benchmark references.

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) **no**. The paper proposes an empirical architecture and does not make formal theoretical contributions.

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) **NA**
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) **NA**
- 2.4. Proofs of all novel claims are included (yes/partial/no) **NA**
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) **NA**
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) **NA**
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) **NA**
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) **NA**

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) **yes**

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) **partial**. The Experiments section motivates synthetic homophily stress tests, Planetoid checks, WebKB, and larger Geom-GCN heterophily benchmarks.
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) **NA**. The paper does not introduce a new fixed dataset.
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **NA**. The paper does not introduce a new fixed dataset.
- 3.5. All datasets drawn from the existing literature (potentially including authors’ own previously published work) are accompanied by appropriate citations (yes/no/NA) **yes**. Existing datasets are cited in the manuscript bibliography.
- 3.6. All datasets drawn from the existing literature (potentially including authors’ own previously published work) are publicly available (yes/partial/no/NA) **yes**. The reported datasets are public and loaded through repository dataset loaders.

- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) [NA](#). [The paper does not rely on non-public datasets](#).

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) [yes](#)

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) [partial](#). [Final settings and selection by validation accuracy are specified in configs; the full development search range is not yet exhaustively documented in the manuscript](#).
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) [partial](#). [Dataset loading and preprocessing code are included in the repository, but not as a separate paper appendix](#).
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) [yes](#). [The repository contains experiment, model, data-loading, summarization, audit, and verification code](#).
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) [partial](#). [The code is prepared for release, but the final public license should be selected before publication](#).
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) [partial](#). [Core model code is documented with class docstrings and clear names; line-by-line links back to paper equations are not yet complete](#).
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) [yes](#). [Seeds are specified in each YAML config and set through the shared set_seed utility](#).
- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) [partial](#). [The repository records software requirements](#)

[and local CPU/Windows context; exact CPU model, memory, and library versions should be frozen before final submission](#).

- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) [partial](#). [Accuracy, standard deviation, and paired tests are described; metric motivation can be expanded in the manuscript](#).
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) [yes](#). [Tables and configs state the number of seeds or fixed splits used for each reported benchmark](#).
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) [yes](#). [Reported tables include standard deviations and paired-split statistical tests where applicable](#).
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) [partial](#). [Paired t-tests are reported for WebKB and larger Geom-GCN; not every diagnostic table has a statistical test](#).
- 4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) [partial](#). [Final hyperparameters are in YAML configs; a compact manuscript or supplementary hyperparameter table should be added before final submission](#).